Zecpath AI Data Entity Design

Day 4: Data Understanding & Structuring | Zecpath AI project | Prepared by Abhishek, Intern

1. Task checklist

Task	Done?	Where
Analyse at least 10 resumes from different domains	Yes, 10 resumes	Section 2
Study technical and non-technical job descriptions	Yes	Section 4
Identify skills, experience patterns, designations, education, certifications	Yes	Section 3
Define Candidate, Job, Skill and Experience entities	Yes	Section 5
Design JSON schema formats for resume and JD	Yes	Sections 6–7

2. The 10 resumes I analysed

#	Domain	Designation	Level	Skills type	Education	Cert.
1	MERN / Full-stack	Full-Stack Developer	Mid (3+ yrs)	Technical	B.Tech CS, VTU, 2022	–
2	Data analyst	Data Analyst	Junior (2+)	Technical	B.Sc. Stats, Loyola, 2023	–
3	UI/UX designer	UI/UX Designer	Mid (3)	Design	B.Des, MIT ID, 2022	–
4	Sales	Senior Sales Executive	Senior (4)	Non-technical	BBA, Amity, 2022	–
5	HR / recruiter	HR Executive – TA	Mid (3)	Non-technical	MBA HR, Rajagiri, 2022	–
6	Digital marketing	Digital Marketing Exec.	Mid (3)	Mixed	B.Com, Christ, 2022	2
7	Accountant	Accountant	Mid (3+)	Non-technical	M.Com, Pune Univ., 2022	–
8	Civil engineer	Site Engineer	Mid (3)	Technical + domain	B.E. Civil, Anna Univ., 2022	–
9	Customer support	Customer Support Exec.	Junior (2.5)	Non-technical	B.A. English, Bangalore Univ., 2022	–
10	Fresher	None (wants Ops / BA)	Fresher	Technical + soft	B.Com, Nagpur Univ., 2026	1

Experience level is my own pick from the title and years. Education: bachelor's in 8, master's in 2 (resumes 5 and 7).

3. What I found

Skills

- Technical roles list tools (React, SQL, Figma); non-technical roles list domain skills (lead generation, Tally, payroll).
- Same skill, different spelling ("React.js" vs "React", "Advanced Excel" vs "MS Excel"), so I keep *name* and add *normalized_name*.
- Some skills appear only in experience text (Redis, JWT, Razorpay, PostgreSQL), so I added *source* and *confidence*.
- Bracket skills like "Python (Pandas, NumPy)" need to be split into separate skills.
- No resume gives skill level or years per skill, so those fields will mostly be null.

Experience patterns

- Dates like "Jan 2024 – Present" become YYYY-MM, with *end_date: null* for the current job.
- Freshers have internships and projects, not jobs (resume 10).
- Calculated experience (about 4.2–4.5 yrs) was higher than the summary (2–4 yrs) in all 9 experienced resumes, so total experience must be calculated from dates ("Present" taken as Nov 2026).
- No gaps or overlaps were found, so I could not test that rule.

Designations

- Titles vary: Junior Web Developer, Graduate Engineer Trainee, Accounts Assistant, HR Coordinator, SEO & Content Associate, Customer Care Associate. So *normalized_designation* is needed. Trainee, Assistant, Coordinator and Associate have no clear level.

Education

- Every resume gives degree, field, institution and end year, but never a start year.
- Only resume 10 gives grades (CGPA 8.4/10, Class XII 89%), so grade is null for the other nine.

Certifications

- Only 2 resumes have them (Google Ads, Google Analytics; Excel Skills for Business). Names only, no dates, so only *name* is required.

Limit: these are clean sample templates; real resumes will be messier.

4. Job descriptions: technical vs non-technical

	Technical (e.g. MERN Developer)	Non-technical (e.g. Sales Executive)
Title	Developer, Engineer, DevOps	Executive, Manager, Associate
Skills	Named technologies, often with min years	Domain and soft skills (CRM, negotiation)
Experience	Years plus seniority	Years plus industry exposure
Education	Usually a related degree	Often any graduate
Certifications	Cloud or vendor certificates	Rare, sometimes domain licences
Extra	Coding / machine test rounds	Communication-heavy screening

Every JD mixes must-have and good-to-have skills, so each job skill gets an *importance* flag. *role_category* decides whether a machine test round is used. Cutoffs, weights, round order and call settings live in the Job profile.

5. Standard data entities

Required fields are marked *.

Entity	Fields
Candidate profile	schema_version*, candidate_id*, personal* (full_name*, email*, phone, location, links), skills*, experience*, education*, summary, total_experience_years, current_designation, certifications, projects, languages, preferences, consent, parsing_meta
Job profile	schema_version*, job_id*, company_id*, title*, role_category*, employment_type*, location*, requirements* (skills, experience, education, certifications, languages), normalized_title, work_mode, compensation, scoring_config, interview_rounds, call_settings, status
Skill object	name*, normalized_name, category, proficiency, years_used, last_used, source, confidence (0–1). In a Job, a skill uses *importance* (must_have / good_to_have) and *min_years* instead of source and confidence.
Experience object	company*, designation*, start_date* (YYYY-MM), end_date (null if current), normalized_designation, employment_type, is_current, duration_months, industry, location, responsibilities, achievements, skills_used

Other entities: **Education** (degree, field_of_study, institution, end_year, grade), **Certification** (name, issuer, issue_date), **Project** (name, description, skills_used), **Language** (language, proficiency). *skills_used* links each job to its skills, so the ATS can check a skill was really used.

6. JSON schema: resume (resume_schema.json)

```
{
 "$schema": "https://json-schema.org/draft/2020-12/schema",
 "title": "Candidate Profile",
 "type": "object",
 "additionalProperties": false,
 "required": ["schema_version", "candidate_id", "personal", "skills", "experience", "education"],
 "properties": {
  "schema_version": {"const": "1.0"},
  "candidate_id": {"type": "string"},
  "personal": {
   "type": "object", "additionalProperties": false,
   "required": ["full_name", "email"],
   "properties": {
    "full_name": {"type": "string"},
    "email": {"type": "string"},
    "phone": {"type": ["string", "null"]},
    "location": {"type": ["string", "null"]},
    "links": {"type": "array", "items": {"type": "object", "required": ["type", "url"],
      "properties": {"type": {"enum": ["github", "linkedin", "kaggle", "behance", "dribbble", "other"]},
                     "url": {"type": "string"}}}}
   }
  },
  "summary": {"type": ["string", "null"]},
  "total_experience_years": {"type": ["number", "null"]},
  "current_designation": {"type": ["string", "null"]},
  "skills": {"type": "array", "items": {"$ref": "#/$defs/skill"}},
  "experience": {"type": "array", "items": {"$ref": "#/$defs/experience"}},
  "education": {"type": "array", "items": {"$ref": "#/$defs/education"}},
  "certifications": {"type": "array", "items": {"type": "object", "required": ["name"],
      "properties": {"name": {"type": "string"}, "issuer": {"type": ["string", "null"]},
                     "issue_date": {"type": ["string", "null"]}}}},
  "projects": {"type": "array", "items": {"type": "object", "required": ["name"],
      "properties": {"name": {"type": "string"}, "description": {"type": ["string", "null"]},
                     "skills_used": {"type": "array", "items": {"type": "string"}}}}}
 },
 "$defs": {
  "skill": {
   "type": "object", "required": ["name"],
   "properties": {
    "name": {"type": "string"},
    "normalized_name": {"type": ["string", "null"]},
    "category": {"enum": ["programming_language", "framework", "database", "cloud_devops", "tool",
                          "design", "business_domain", "soft_skill", "other"]},
    "proficiency": {"enum": ["beginner", "intermediate", "advanced", "expert", null]},
    "years_used": {"type": ["number", "null"]},
    "source": {"enum": ["skills_section", "experience", "project", "certification", "inferred"]},
    "confidence": {"type": "number", "minimum": 0, "maximum": 1}
   }
  },
  "experience": {
   "type": "object", "required": ["company", "designation", "start_date"],
   "properties": {
    "company": {"type": "string"},
    "designation": {"type": "string"},
    "normalized_designation": {"type": ["string", "null"]},
    "employment_type": {"enum": ["full_time", "part_time", "internship", "contract", "freelance", null]},
    "start_date": {"type": "string", "pattern": "^[0-9]{4}-[0-9]{2}$"},
    "end_date": {"type": ["string", "null"], "pattern": "^[0-9]{4}-[0-9]{2}$"},
    "is_current": {"type": "boolean"},
    "responsibilities": {"type": "array", "items": {"type": "string"}},
    "skills_used": {"type": "array", "items": {"type": "string"}}
   }
  },
  "education": {
   "type": "object", "required": ["degree", "institution"],
   "properties": {
    "degree": {"type": "string"},
    "field_of_study": {"type": ["string", "null"]},
    "institution": {"type": "string"},
    "end_year": {"type": ["integer", "null"]},
    "grade": {"type": ["object", "null"],
              "properties": {"value": {"type": ["number", "null"]}, "scale": {"type": "string"}}}
   }
  }
 }
}
```

7. JSON schema: job description (job_schema.json)

```
{
 "$schema": "https://json-schema.org/draft/2020-12/schema",
 "title": "Job Profile",
 "type": "object",
 "additionalProperties": false,
 "required": ["schema_version", "job_id", "company_id", "title", "role_category",
              "employment_type", "location", "requirements"],
 "properties": {
  "schema_version": {"const": "1.0"},
  "job_id": {"type": "string"},
  "company_id": {"type": "string"},
  "title": {"type": "string"},
  "normalized_title": {"type": ["string", "null"]},
  "role_category": {"enum": ["technical", "non_technical"]},
  "employment_type": {"enum": ["full_time", "part_time", "internship", "contract", "freelance"]},
  "work_mode": {"enum": ["onsite", "remote", "hybrid", null]},
  "location": {"type": "object", "required": ["city", "country"],
               "properties": {"city": {"type": "string"}, "state": {"type": ["string", "null"]},
                              "country": {"type": "string"}}},
  "requirements": {
   "type": "object", "required": ["skills"],
   "properties": {
    "skills": {"type": "array", "items": {"type": "object", "required": ["name", "importance"],
       "properties": {"name": {"type": "string"}, "normalized_name": {"type": ["string", "null"]},
                      "category": {"type": "string"},
                      "importance": {"enum": ["must_have", "good_to_have"]},
                      "min_years": {"type": ["number", "null"]}}}},
    "experience": {"type": "object",
       "properties": {"min_years": {"type": "number"}, "max_years": {"type": ["number", "null"]},
                      "level": {"type": ["string", "null"]}}},
    "education": {"type": "array", "items": {"type": "string"}},
    "certifications": {"type": "array", "items": {"type": "string"}},
    "languages": {"type": "array", "items": {"type": "string"}}
   }
  },
  "compensation": {"type": "object",
     "properties": {"min": {"type": ["number", "null"]}, "max": {"type": ["number", "null"]},
                    "currency": {"type": "string"}, "period": {"type": "string"},
                    "is_disclosed": {"type": "boolean"}}},
  "scoring_config": {"type": "object",
     "properties": {"ats_cutoff_score": {"type": "number"}, "weights": {"type": "object"}}},
  "interview_rounds": {"type": "array", "items": {"type": "object",
     "properties": {"round_number": {"type": "integer"}, "round_type": {"type": "string"},
                    "cutoff_score": {"type": "number"}, "weightage": {"type": "number"}}}},
  "call_settings": {"type": "object",
     "properties": {"voice_gender": {"type": "string"}, "language": {"type": "string"},
                    "max_retries": {"type": "integer"}, "retry_after_minutes": {"type": "integer"}}},
  "status": {"type": "string"}
 }
}
```

8. Sample data

Candidate (from resume 1)

```
{
 "schema_version": "1.0",
 "candidate_id": "C-0001",
 "personal": {"full_name": "Rohan Sharma", "email": "rohan.dev@email.com", "phone": "+91 98765 43210",
              "location": "Bengaluru, India",
              "links": [{"type": "github", "url": "github.com/rohandev"},
                        {"type": "linkedin", "url": "linkedin.com/in/rohandev"}]},
 "total_experience_years": 4.4,
 "current_designation": "Full-Stack Developer",
 "skills": [{"name": "React.js", "normalized_name": "react", "category": "framework", "source": "skills_section"},
            {"name": "Redis", "normalized_name": "redis", "category": "database", "source": "experience"}],
 "experience": [{"company": "TechNova Solutions", "designation": "Full-Stack Developer",
                 "start_date": "2024-01", "end_date": null, "is_current": true,
                 "skills_used": ["react", "node.js", "redis"]},
                {"company": "PixelCraft Digital", "designation": "Junior Web Developer",
                 "start_date": "2022-06", "end_date": "2023-12", "is_current": false}],
 "education": [{"degree": "B.Tech", "field_of_study": "Computer Science",
                "institution": "Visvesvaraya Technological University", "end_year": 2022, "grade": null}]
}
```

Job (MERN Developer)

```
{
 "schema_version": "1.0",
 "job_id": "J-0001",
 "company_id": "CO-01",
 "title": "MERN Developer",
 "role_category": "technical",
 "employment_type": "full_time",
 "location": {"city": "Bengaluru", "state": "Karnataka", "country": "India"},
 "requirements": {
  "skills": [{"name": "React", "importance": "must_have", "min_years": 2},
             {"name": "Docker", "importance": "good_to_have", "min_years": null}],
  "experience": {"min_years": 2, "max_years": 5, "level": "mid"}
 },
 "scoring_config": {"ats_cutoff_score": 60, "weights": {"skills": 0.6, "experience": 0.3, "education": 0.1}},
 "interview_rounds": [{"round_number": 1, "round_type": "hr", "cutoff_score": 50, "weightage": 0.3},
                      {"round_number": 2, "round_type": "technical", "cutoff_score": 60, "weightage": 0.7}]
}
```

9. Design rules I followed

- Every schema has *schema_version*; dates are YYYY-MM; categories use fixed enums.
- Few required fields, because resumes are messy. Missing means null or empty, never invented.
- Original and normalized values stored together.
- No gender, age, photo or marital status fields, so scoring stays unbiased.
- *additionalProperties: false* on top-level objects to catch typos.

10. For team lead review and next steps

- Add dates to Project; add an achievements list; add methodology to skill categories.
- Decide if internships count in total_experience_years.
- Rules JSON Schema cannot check (weights add up to 1.0, end_date after start_date) will be done in Python.
- Next: upgrade the parser to match the schema, build a skill dictionary and a designation map.